

WEEK – 2 PRACTICAL

```
mysql> CREATE TABLE IF NOT EXISTS loans (  
->   loan_id INT AUTO_INCREMENT PRIMARY KEY,  
->   member_id INT NOT NULL,  
->   book_id INT NOT NULL,  
->   loan_date DATE DEFAULT (CURRENT_DATE),  
->   return_date DATE NULL,  
->  
->   FOREIGN KEY (member_id) REFERENCES members(member_id),  
->   FOREIGN KEY (book_id) REFERENCES books(book_id)  
-> );  
ERROR 1046 (3D000): No database selected  
mysql> USE bookflow_db;  
Database changed  
mysql> CREATE TABLE IF NOT EXISTS loans (  
->   loan_id INT AUTO_INCREMENT PRIMARY KEY,  
->   member_id INT NOT NULL,  
->   book_id INT NOT NULL,  
->   loan_date DATE DEFAULT (CURRENT_DATE),  
->   return_date DATE NULL,  
->  
->   FOREIGN KEY (member_id) REFERENCES members(member_id),  
->   FOREIGN KEY (book_id) REFERENCES books(book_id)  
-> );  
Query OK, 0 rows affected (0.11 sec)  
  
mysql> INSERT INTO loans (member_id, book_id)  
-> VALUES  
-> (1, 1),  
-> (2, 2);  
Query OK, 2 rows affected (0.02 sec)  
Records: 2  Duplicates: 0  Warnings: 0  
  
mysql> SELECT  
->   m.full_name AS Member_Name,  
->   b.title AS Book_Title  
-> FROM loans l  
-> JOIN members m ON l.member_id = m.member_id  
-> JOIN books b ON l.book_id = b.book_id  
-> WHERE l.return_date IS NULL;
```

```
+-----+-----+  
| Member_Name | Book_Title |  
+-----+-----+  
| Anil Kumar  | The Alchemist |  
| Priya Sharma | Clean Code  |  
+-----+-----+  
2 rows in set (0.00 sec)  
  
mysql> SELECT  
->   published_year,  
->   COUNT(book_id) AS total_books  
-> FROM books  
-> GROUP BY published_year  
-> ORDER BY published_year;  
+-----+-----+  
| published_year | total_books |  
+-----+-----+  
| 1988          | 1          |  
| 2008          | 1          |  
| 2018          | 1          |  
+-----+-----+  
3 rows in set (0.07 sec)  
  
mysql> COUNT(book_id)  
-> GROUP BY published_year  
-> ALTER TABLE books  
-> ADD COLUMN category VARCHAR(100);  
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'C  
OUNT(book_id)  
GROUP BY published_year  
ALTER TABLE books  
ADD COLUMN category VAR' at line 1  
mysql> CREATE TABLE IF NOT EXISTS donation_history (  
->   donation_id INT AUTO_INCREMENT PRIMARY KEY,  
->   book_id INT NOT NULL,  
->   donor_name VARCHAR(100) NOT NULL,  
->   category VARCHAR(100) NOT NULL,  
->   donation_date DATE DEFAULT (CURRENT_DATE),  
->   FOREIGN KEY (book_id) REFERENCES books(book_id)  
-> );  
Query OK, 0 rows affected (0.08 sec)
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO books (title, isbn, published_year, category)
-> VALUES ('The Psychology of Money', '9780857197689', 2020, 'Finance');
ERROR 1054 (42S22): Unknown column 'category' in 'field list'
mysql> INSERT INTO donation_history
-> (book_id, donor_name, category)
-> VALUES
-> (LAST_INSERT_ID(), 'Rahul Kumar', 'Finance');
Query OK, 1 row affected (0.03 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.05 sec)

mysql> SELECT * FROM books;
+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist  | 9780061122415 | 1988           |
| 2       | Clean Code    | 9780132350884 | 2008           |
| 3       | Atomic Habits | 9780735211292 | 2018           |
+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql> SELECT * FROM donation_history;
+-----+-----+-----+-----+-----+
| donation_id | book_id | donor_name | category | donation_date |
+-----+-----+-----+-----+-----+
| 1           | 1       | Rahul Kumar | Finance  | 2026-08-17    |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> ROLLBACK;
Query OK, 0 rows affected (0.00 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO books (title, isbn, published_year, category)
```

```
ERROR 1054 (42S22): Unknown column 'category' in 'field list'
mysql>
mysql> INSERT INTO donation_history
-> (book_id, donor_name, category)
-> VALUES (9999, 'Test Donor', 'Testing');
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails ('bookflow_db`.`donation_history`, CONSTRAINT `donation_history_ibfk_1` FOREIGN KEY (`book_id`) REFERENCES `books` (`book_id`))
mysql>
mysql> ROLLBACK;
Query OK, 0 rows affected (0.01 sec)

mysql> BEGIN/START TRANSACTION
->
-> Insert Book
->
-> Insert History
->
-> Both successful?
->
-> YES          NO
->
-> COMMIT      ROLLBACK
-> CREATE INDEX idx_books_isbn
-> ON books(isbn);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near '/'
START TRANSACTION
?
Insert Book
?
Insert History
' at line 1
mysql> SELECT *
-> FROM books
-> WHERE isbn = '9780061122415';
+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist  | 9780061122415 | 1988           |
+-----+-----+-----+-----+
1 row in set (0.05 sec)
```

```
mysql> SELECT *
-> FROM books
-> WHERE title = 'The Alchemist';
+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist  | 9780061122415 | 1988           |
+-----+-----+-----+-----+
1 row in set (0.00 sec)
```